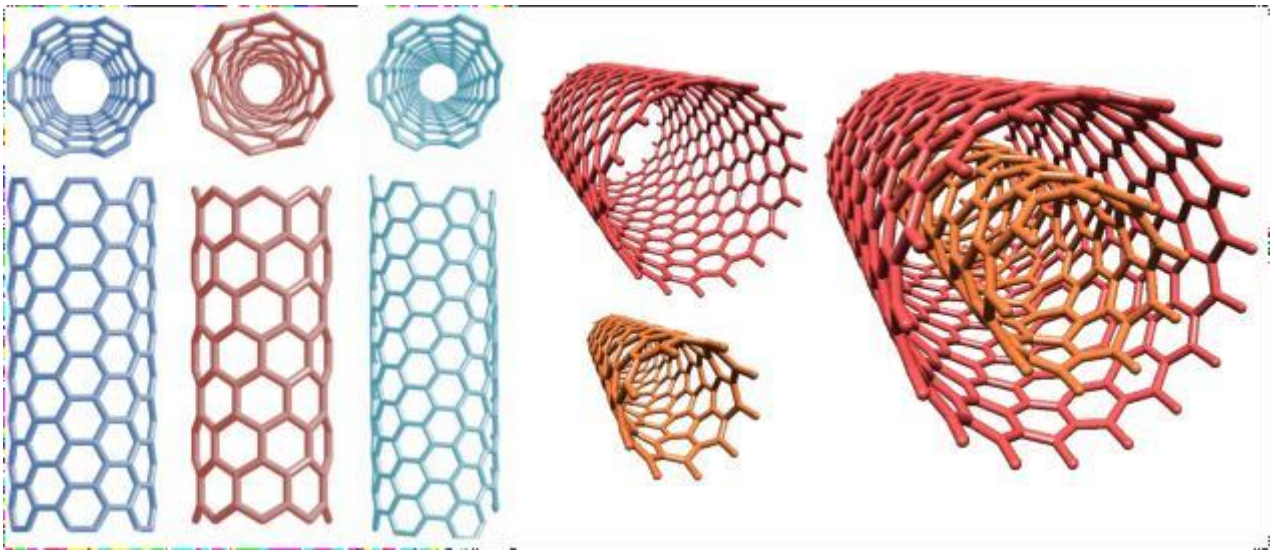
UNIT-IV: CHEMISTRY OF NANOMATERIALS



UNIT-IV

TITLE OF THE UNIT- CHEMISTRY OF NANOMATERIALS

Abstract

Nanomaterials: Size dependent properties, Classification and Synthesis

The properties of nanomaterials can be significantly different from those of the same material in bulk form. This is due to the large surface area to volume ratio of nanomaterials, which can lead to changes in their optical, electrical, mechanical, and catalytic properties. Nanomaterials can be classified based on their size, shape, and composition. Nanoparticles are typically less than 100 nanometers in size, while nanotubes and nanofibers are typically less than 100 nanometers in diameter. Nanoparticles can be spherical, cubic, or rod-shaped, and they can be made of a variety of materials, including metals, semiconductors, and polymers. There are a variety of methods for synthesizing nanomaterials. One common method is solution combustion, which involves burning a metal salt in a solution to produce nanoparticles. Another common method is sol-gel synthesis, which involves the hydrolysis and condensation of metal alkoxides to produce nanoparticles.

Synthesis and Applications of Carbon Nanotubes and Graphenes

Carbon nanotubes and graphene are two types of nanomaterials that have attracted significant interest due to their unique properties. Carbon nanotubes are long, thin tubes of carbon atoms that can be either single-walled or multi-walled. Graphene is a single layer of carbon atoms arranged in a hexagonal lattice. Carbon nanotubes and graphene have a wide range of potential applications, including in electronics, energy, and catalysis. For example, carbon nanotubes can be used to make high-performance transistors and batteries. Graphene can be used to make high-strength composites and membranes.

Nano Lubricants

Nano lubricants are lubricants that contain nanoparticles. Nanoparticles can improve the lubricating properties of fluids by filling in surface roughness and reducing friction. Nano lubricants have potential applications in a variety of industries, including defense, automotive, and aerospace.

This syllabus provides an overview of the size dependent properties, classification, synthesis, and applications of nanomaterials. The syllabus is designed for students who are interested in learning more about this rapidly developing field.

Blow-up Syllabus

Sl No	Topic to be taught	Duration
1	Introduction to nanomaterials -Classification	1 hour
2	Size dependent properties of nanomaterials- Surface area, optical and catalytic properties.	1 hour
3	Synthesis of nanomaterials- Combustion and sol-gel methods	1 hour
4	Carbon Nanomaterials-Introduction, synthesis and properties	1 hour
5	Carbon Nanomaterials- functionalization and application	1 hour
6	Graphene-Introduction, synthesis, and properties	1 hour
7	Graphene- functionalization and application	1 hour
8	Nano lubricants –Different types of nanoparticles as lubricant additives	1 hour
9	Application of Nano lubricants- in defence, automobile and space-crafts.	1 hour

Introduction

Nanomaterials are defined as materials with an average grain size less than 100 nanometers. The word "nano" originates from the Greek word "nanos" which means "dwarf". However, in scientific language it is a prefix which has a value equal to "one billionth, i.e. 10^{-9} Therefore, one nanometer is one billionth of a meter (1 nm = 10^{-9} m).

Examples:

1. Diameter of a hydrogen atom is about 0.1 nm. If ten hydrogen atoms are aligned in a line, then the resulting length would be approximately 1 nm. Similarly, 1 nm is approximately 5 silicon atoms aligned in a line.
2. A single strand of human hair is around 20,000 nm in diameter. A nanometer is one millionth of a millimeter - approximately 100,000 times smaller than the diameter of a human hair.
3. Size of a DNA molecule is about 2.5 nm.

4.2 Properties of Nanomaterials

Nanomaterials have the structural features in between of atoms /molecules and the bulk materials. The properties of materials with nanometer dimensions are significantly different from those of atoms and bulks materials. This is mainly due to the nanoscale dimension of the materials which render them: ***(i) large fraction of surface atoms (ii) high surface energy*** (iii) ***spatial confinement*** and (iv) ***reduced imperfections***, which do not exist in the corresponding bulk materials.

Due to their small dimensions, nanomaterials have extremely ***large surface area to volume ratio***, which makes a large to be the surface or interfacial atoms, resulting in more “surface” dependent material properties.

Surface area:

As particle dimensions reduce towards the nanoscale, the surface-to-volume ratio proportionately increases. It further increases the surface active cites (reaction spots), which is the main reason for changing the properties of nanomaterial.

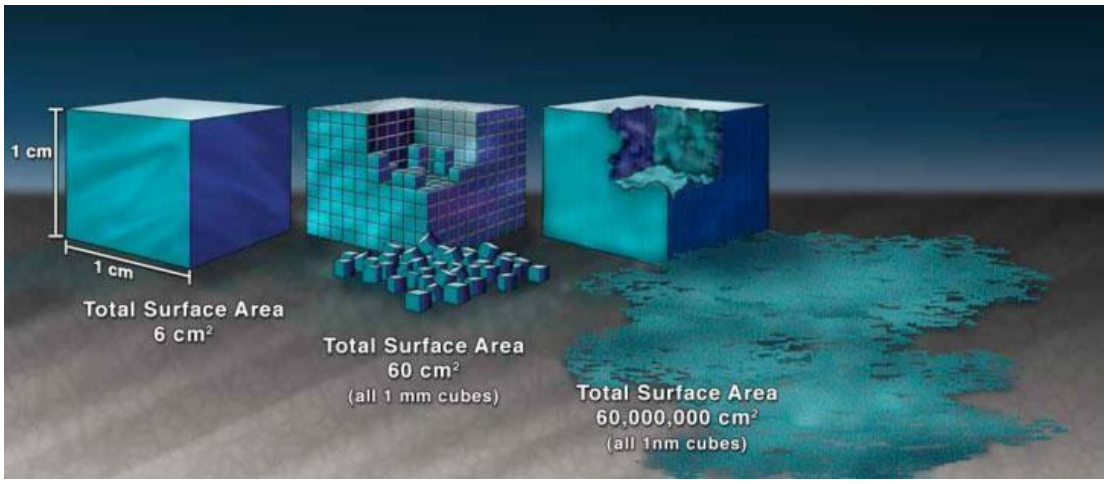


Fig. 4.1: Surface area variation from bulk to nano.

Optical Properties

Optical properties of nanomaterials can be significantly different from their bulk counterpart. The change in optical properties is mainly due to the **surface plasmon resonance and the increased energy level spacing.**

Surface Plasmon Resonance (SPR): The coherent excitation of entire free electrons in the conduction band may produce an in-phase oscillation, when the size of a metal nanocrystal is smaller than the wavelength of incident radiation, **called surface Plasmon resonance.** Thus, plasmon resonance depends on the particle size.

In solid state physics, the plasmon represents the **collective oscillation of a free charge** in a metal, and may be considered as a kind of **plasma wave**. The positive electrical charge in the metal **is fixed** and the free electron **is free to move** around it. An applied external electric field, as from a light source, causes the free electrons at the surface of the metal to vibrate collectively, giving rise to surface plasmons. "Since electrons are also particles with an electric charge, when they vibrate they also generate an electric field, and when the electric field from the vibration of free electrons and the applied external electric field (e.g., electromagnetic waves) resonate the resulting phenomenon is referred to as a **surface plasmon resonance"** that takes place at the surface of the metal.

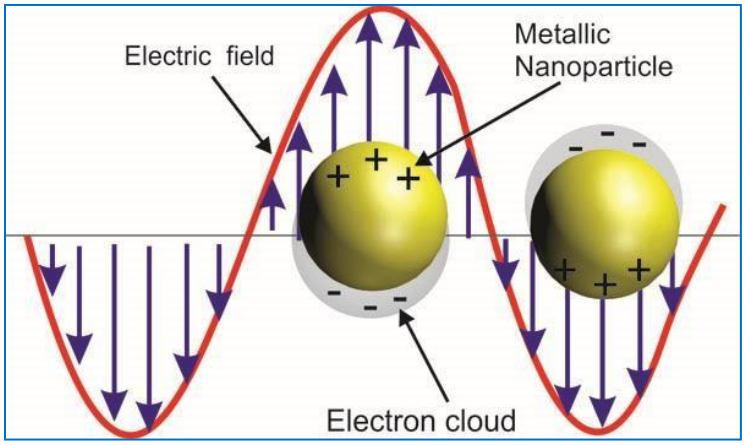


Fig. 4. 2: SPR phenomena

For example, the wavelength of the SPR band maximum of a spherical **Au nanoparticle is 520–550 nm**. If a colloidal Au nanoparticle solution is now irradiated with visible light at these wavelengths (520–550 nm), the visible light corresponding to the **green color** is absorbed and the particles now display a **red purple** color, which is the **complementary color to green**. In a colloidal **Ag nanoparticle** solution which has a SPR band maximum near **400 nm**, the **blue color** of the visible light is absorbed and the Ag particles now take on **a yellow color**, the complementary color to blue.

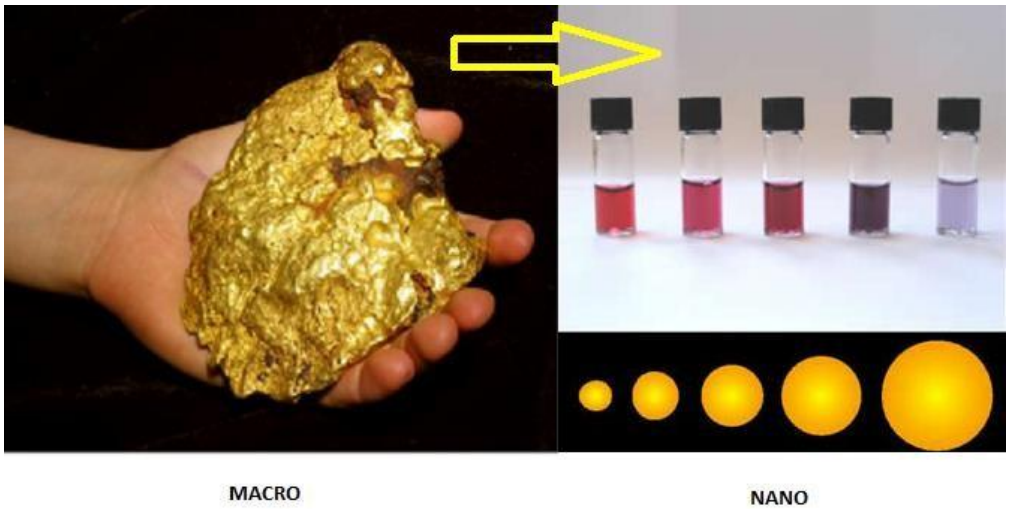


Fig. 4.3: Fluorescence emission of (a) Gold nanoparticles, and (b) Silver nanoparticles and of various sizes and absorbance

Catalytic Properties of Nanomaterials

Nanoscale catalysts have high specific surface area and surface energy, which ultimately lead to the high catalytic activity. Nano-catalyst improves the selectivity of the reactions by allowing reaction at a lower temperature, reducing occurrence of side reactions, higher recycling rates and recovery of energy consumption. Therefore, these are widely used in green chemistry, environmental remediation, efficient conversion of biomass, renewable energy development and other areas of interest.

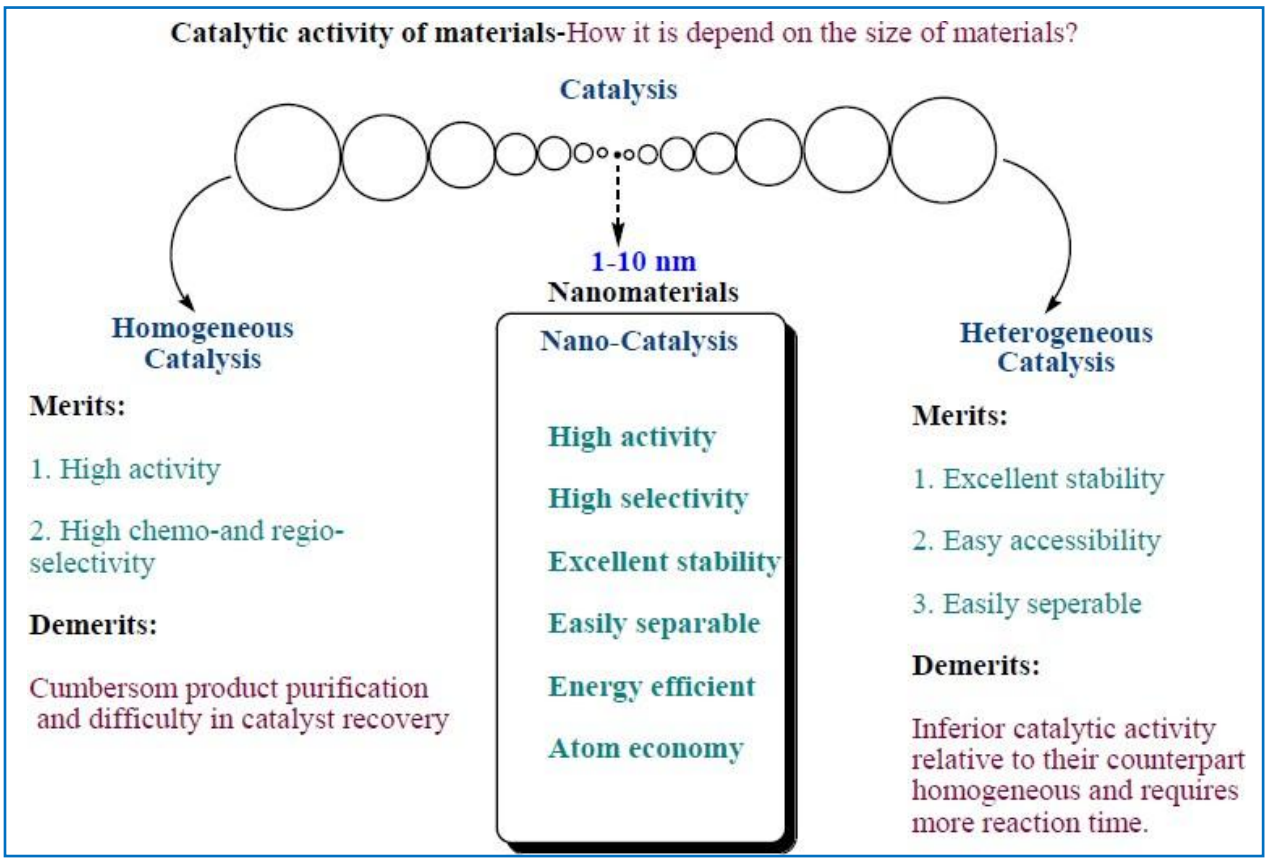
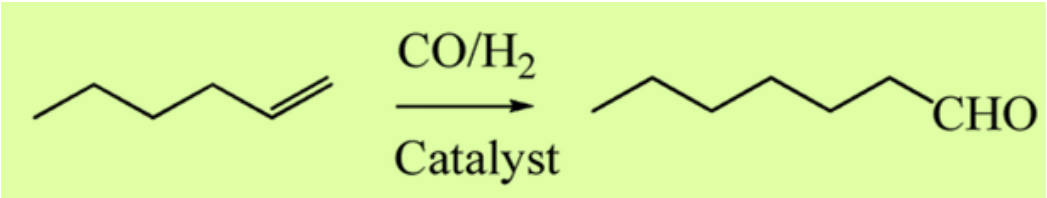


Fig. 4.4: Properties of Nano-catalyst

The rate of any chemical reaction depends on the number of active reaction sites on the catalyst. The surface area of nanoscaled catalyst is larger than that of bulk material. As compared to bulk catalyst, the nanoscale catalyst of same material with same quantity has more number of active sites. Because of unsaturated valencies of surface atoms of Nanoparticle catalyst, they catalyze the reactions at much faster rates due to higher surface energies as compared to their respective bulk partners. Therefore, the nanoscaled material is found to exhibit very high catalytic efficiency than the corresponding bulk material.

Ex: 1. ***Hydroformylation of 1–Hexene over Rh/Nano -oxide***



The highest activity with 100% total conversion and 96% yield of aldehydes was obtained with the Rh/nano-ZnO catalyst.

The Rh/nano-ZnO catalyst was found to be more stable and active than the corresponding rhodium catalyst supported on bulk ZnO (76% yield).

Article

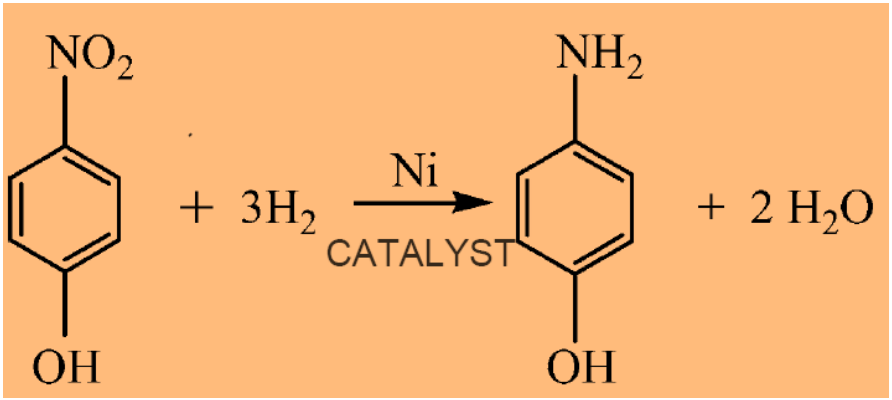
Hydroformylation of 1-Hexene over Rh/Nano-Oxide Catalysts

Maija-Liisa Kontkanen [1], Matti Tuikka [1,2], Niko M. Kinnunen [1], Sari Suvanto [1] and Matti Haukka [1,2,*]

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2. *Catalytic hydrogenation of p-nitrophenol to p-aminophenol using nano-sized nickel*



- *The hydrogenation rate of nano-sized nickel is about 15 times higher than that of Raney Ni at similar reaction conditions*
- *The **nano-sized nickel** system suffers 54% deactivation, while the **Raney nickel** system suffers 80.4% deactivation only throughout six continuous hydrogenation cycles.*
- *These results indicate that the catalytic stability of **nano-sized nickel** system is superior to that of the **Raney nickel system***

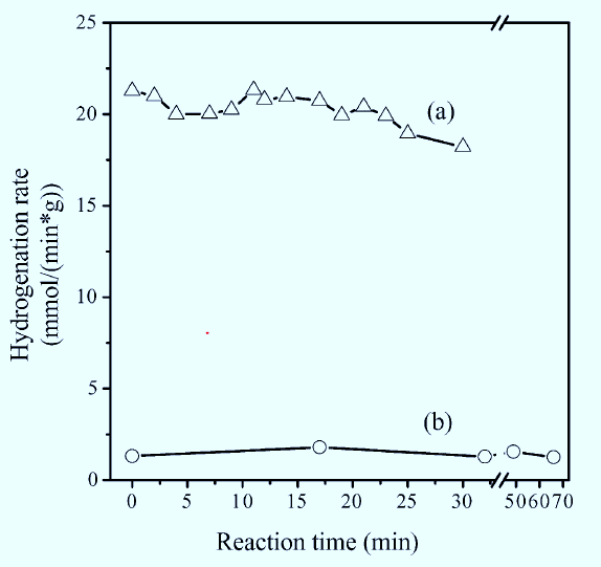


Fig. 4.5: Comparison of nano and bulk Ni catalyst property

Applied Catalysis A: General 277 (2004) 259–264

Synthesis of *p*-aminophenol from *p*-nitrophenol over nano-sized nickel catalysts

Yan Du, Hongling Chen*, Rizhi Chen, Nanping Xu

4.3 Classification of Nanomaterials

Nanostructured materials are classified as

1. Zero dimensional (0D): Nanomaterials are less than 100 nm in all three dimensions.
2. One dimensional (1D): Nanomaterials are less than 100 nm in any two dimensions.
3. Two dimensional (2D): Nanomaterials are less than 100 nm in any one dimensions, and
4. Three dimensional (3D) nanostructures: Nanomaterials are aggregations of any of the above (0, 1, 2D) dimensional nanomaterial with greater than 100nm in all the three

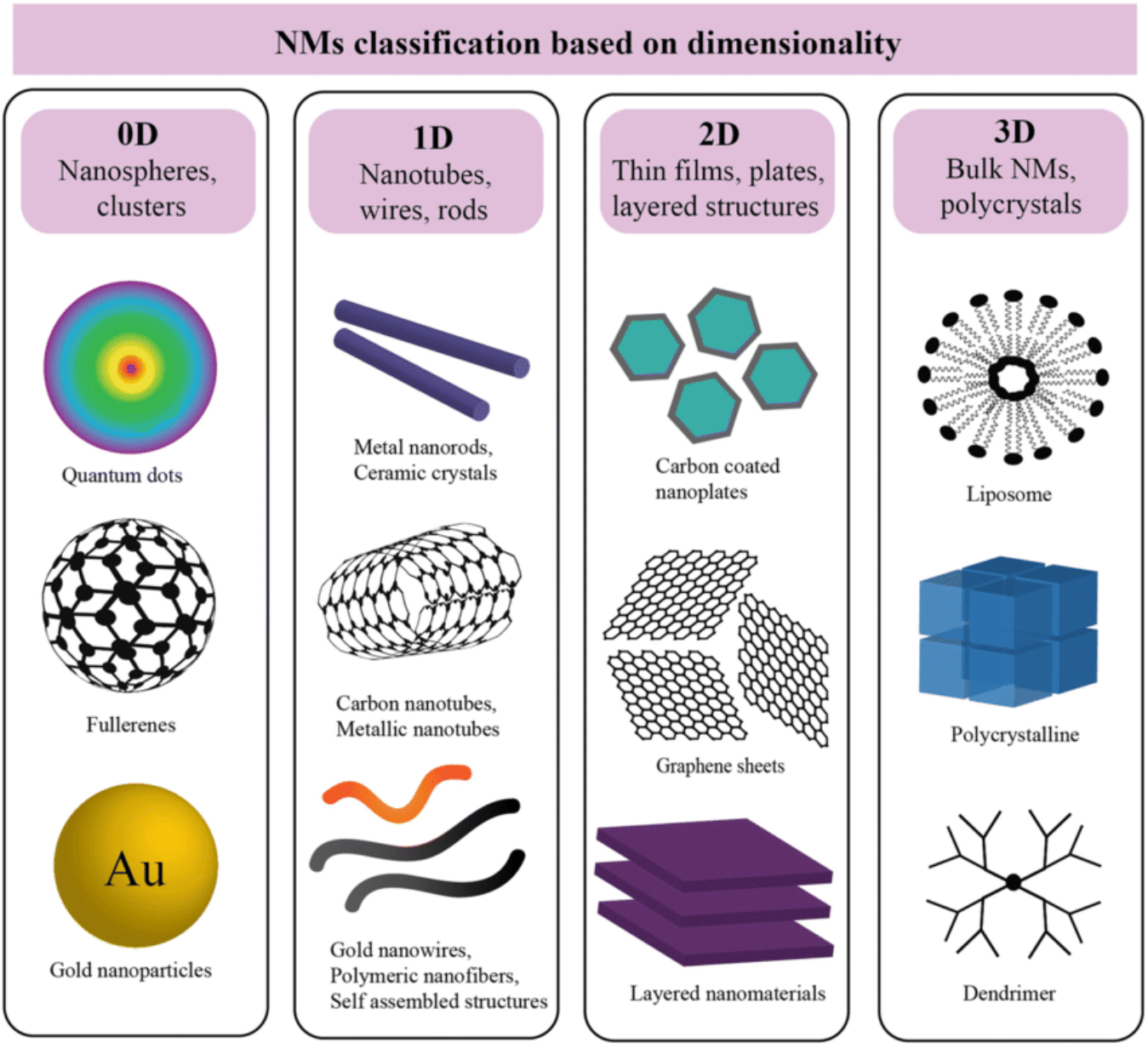


Fig. 4.6: Examples for types nanomaterials

4.4 Synthesis of Nanomaterials:

1. Low temperature solution combustion (SCS) method

Combustion method: is a low temperature, time saving, energy efficient, self-propagating method, involving **spontaneous exothermic redox reaction** between **metal nitrate and organic fuel**, used to prepare nanomaterials.

Principle: The method is based on the utilization of heat energy produced during the exothermic spontaneous redox reaction between an oxidizer (metal nitrate) and a reducing agent (organic fuel). The oxidizer can be of any metal nitrates and reducing agents may be organic fuels, such as glycine, oxalic acid, urea, hexamine, sugar, EDTA, Dextrose etc.

$$\text{Metal nitrate + Fuel} \xrightarrow{500\ ^{\circ}C} \text{Nano Metal oxide + Gases}$$

The amount of fuel can be calculated in such way that the total oxidizing valency of fuel should match with the total reducing valency of metal nitrate. The following formula can be used to calculate the amount of fuel:

$$\frac{\text{Wt. of fuel}}{\text{Mol. Wt. of fuel}} \times \text{Valency of fuel} = \frac{\text{Wt. of metal nitrate}}{\text{Mol. Wt. of metal nitrate}} \times \text{Valency of metal nitrate}$$

For the calculation of valency of fuel and metal nitrate, the valency to be considered for the elements such as of N, O, H, C, and M^{n+} are 0, -2, +1, +4 and +n respectively.

Procedure:

Step1: Weighing and mixing of the reactants:



Weighing of all the reactants (metal nitrate and a fuel) according to stoichiometry, by using the above formula and dissolved in minimum quantity of distilled water in a beaker of suitable size.

The mixture is stirred on a magnetic stirrer till to get uniform solution as shown in figure (left side).

Magnetic stirrer on which uniform redox solution is obtained by stirring

Fig. 4.7: Magnetic stirrer

Step 2: Evaporation and combustion:

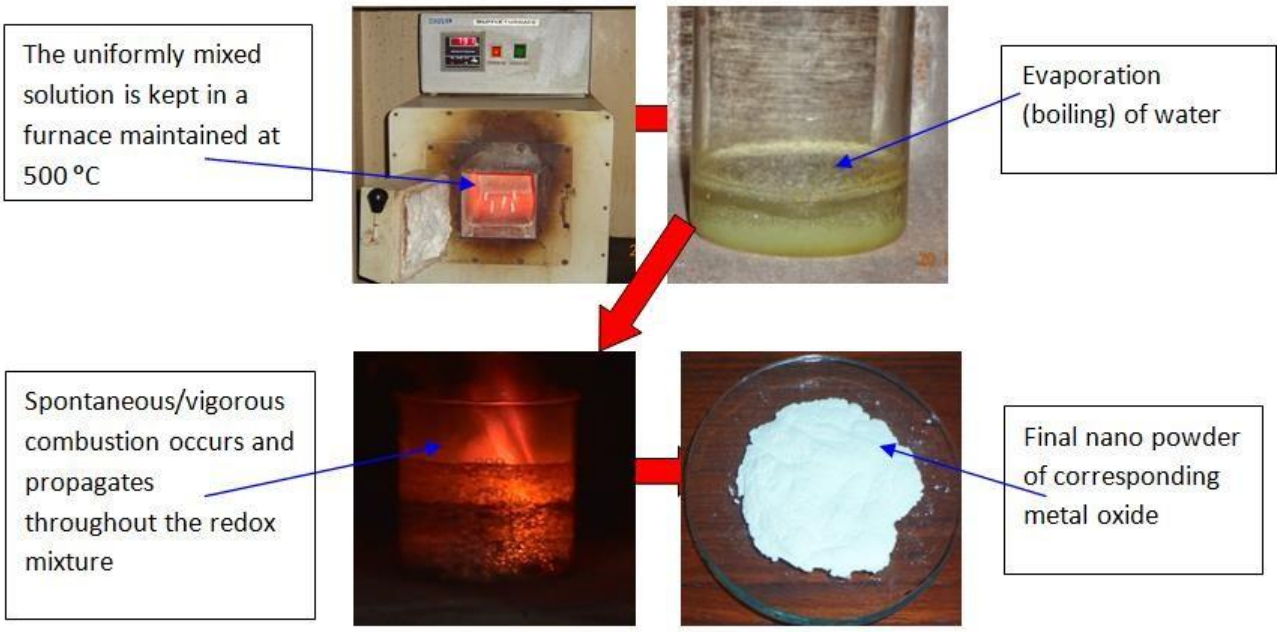


Fig. 4.8: Solution combustion syntheses steps

The clear solution is kept in a furnace (shown below) maintained at 500 ºC. Firstly the solution undergoes evaporation that results in more concentrated, uniformly mixed viscous-gel type substance. After some time, the viscous-gel catches fire and propagates spontaneously in the redox mixture in the form of either a flame type or smouldering type. The combustion lasts for about 1-2 min. During the flame propagation large quantity of gasses and high temperature produced helps in the formation of respective nano metal oxide. The SCS has emerged as a viable technique for the preparation of advanced nano-metal oxide.

The **most important advantages of SCS process are:**

(i) It is an easy and fast process that uses relatively simple equipment.

(ii) High-purity products can be easily prepared by using this method.

(iii) Composition, structure, homogeneity, size and stoichiometry of the products can be controlled.

(iv) High exothermicity of the metal nitrate–fuel reaction permits incorporation of desired quantity of dopants in the various hosts of industrially useful phosphors.

(v) This is the cheap method as compared to conventional solid state method.

Example: Synthesis of MgO by combustion method:

Chemicals: Magnesium nitrate ($Mg(NO_3)_2.6H_2O$, (Mol. Wt=256.41) any organic fuel (say urea; NH_2CONH_2; Mol. Wt = 60) and distilled water.

Instruments/Glassware: High Temperature Furnace, 250ml Borosil glass beaker, Tong, Spatula, Weighing balance, Magnetic stirrer.

Calculation of fuel:

The valency of urea, N_2H_4CO= 2(0)+4(+1)+1(+4)+1(-2)= 6

The valency of $Mg(NO_3)_2.6H_2O$= 1(+2)+2(0)+ 6(-2)+6(0)= 10

The amount of urea for 5.0 g of $Mg(NO_3)_2$ is calculated as follows.

$$\frac{\text{Wt. of urea}}{\text{Mol. Wt. of urea}} \times \text{Valency of urea} = \frac{\text{Wt. of Mg-nitrate}}{\text{Mol. Wt. of Mg-nitrate}} \times \text{Valency of Mg-nitrate}$$

Wt of urea = (5x10x60)/(256.41x6)= 1.95g

Procedure: weigh exactly 5.0g of Magnesium nitrate and 1.95 g of urea (1:1 stoichiometry) in 250 ml beaker and dissolve the salts by adding minimum quantity of distilled water and stirring on a magnetic stirrer for 10 min. The uniformly dissolved redox mixture is kept in a pre-heated furnace maintained at 500 ºC. The redox solution undergoes evaporation and finally results in viscous-gel containing uniformly mixed magnesium nitrate and urea. After some time it catches fire with the liberation of gases and heat energy, finally results in highly porous, white powder of nano MgO. The nano characteristics formed powder is further confirmed by using powder X-ray diffraction technique (crystalline phase), Transmission Electron microscope (nano size).

Exercise:

Calculate the wt of fuel and write the combustion procedure for the synthesis of

1) ZnO from $Zn(NO_3)_2.6H_2O$ (Mol. Wt. =297.49) using glycine (NH_2CH_2COOH, Mol. Wt =75)

2) CaO from $Ca(NO_3)_2.4H_2O$ (Mol. Wt. =236.15) using glucose ($C_6H_{12}O_6$= Mol. Wt =180)

^^^

The reactions involved in the sol-gel process are

1. Hydrolysis

$$M\text{-}OR + H_2O \longrightarrow M\text{-}OH + ROH$$

Metal alkoxide Metal hydroxide

2. Condensation

$$M\text{-}OR + H\ O\text{-}M \longrightarrow M\text{-}O\text{-}M + R\text{-}OH$$

Condensation…… Condensation …….. Condensation ……of …..M-O-M results in to a polymer network in all the possible directions.

Process:

Step 1: Preparation of homogeneous solution either by dissolution of metal salt in water or metal alkoxide in an organic solvent (usually alcohol) that is miscible in water.

Step 2: 'Sol' formation by hydrolysis: It involves the conversion of homogeneous solution into a "sol". (A stable dispersion of colloidal particles of precursors in a solvent is known as sol) due to hydrolysis reaction. During hydrolysis, the alkoxide groups (-OR) are replaced via the nucleophilic attack of the oxygen atom of a water molecules, results in release of alcohol and formation of metal hydroxide (sol).

Step 3: 'Gel' formation by condensation

The colloidal solution is kept for aging. During aging condensation reaction between two metal hydroxyl/alkoxy species leads to M-O-M bonds with the release of H_2O/R-OH. This condensation process continues and finally results in a "gel", an interconnected, a rigid and porous inorganic network covered completely with liquid phase. This transformation is called **Sol-Gel transition.**

Step 4: Drying of gel:

It involves removal of liquid phase from gel network. There are different ways of drying gel:

(a) If the gel medium is dried, by removing the liquid solvent (under hyper critical conditions) without destroying the gel network, **"aerogel" is** produced.

(b) If the solvent is dehydrated by under ambient conditions (removal of R-OH groups), **"xerogel"** are produced.

(c) If the gel network is sintered at high temperature (800ºC), densification, decomposition of gel results in complete collapse of gel net into powder.

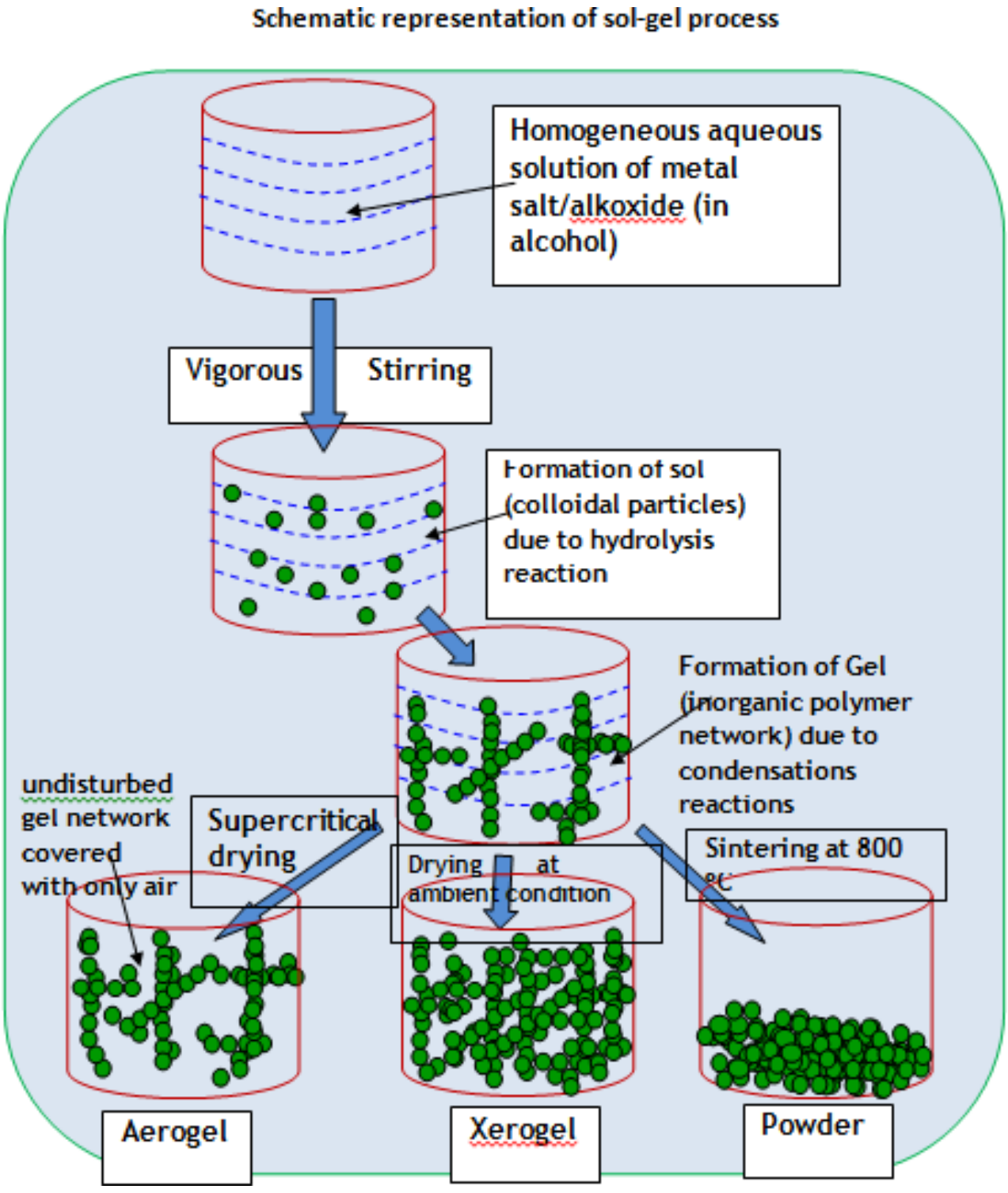


Fig. 4.9: Sol-gel synthesis flowchart

Example: Preparation of TiO_2 by Sol-gel method:

Chemicals required: Titanium isopropoxide [$Ti(O(CH(CH_3)_2)_4$], absolute alcohol, distilled water, HNO_3

Equipment/glassware:

i) 250 ml glass beaker, magnetic stirrer, Furnace, oven.

Procedure: Stoichiometric quantity of titanium (IV) isopropoxide is dissolved in absolute ethanol and distilled water in the ratio of Ti:H_2O=1:4. HNO_3 is added to adjust pH and restrain the hydrolysis process of the solution.

ii) The solution is vigorously stirred for 30 min in order to form sol (a suspension of colloidal particle).

iii) The sol is kept for aging for about 24 h. During aging sol is transformed into gel. (It is due to polycondensation and formation of inorganic network).

iv) In order to obtain TiO_2 nanoparticles, the gels are dried at 120 ºC for 2 h to evaporate water and organic solvent. Then the dry gel is sintered at 450 ºC for 2 h to obtain TiO_2 nanoparticles.

Reaction mechanism: Nucleophilic (H_2O) attack on alkoxy group.

The advantages of this technique are:

1. It is a simple and economic technique, as the fabrication does not need expansive machinery or apparatuses.
2. It offers a high flexibility as one can produce materials with a wide range of stoichiometry and additional dopants.
3. It allows the fabrication of high quality coatings.
4. The starting materials are easily to obtain, not expensive and available in a high purity.
5. Nanomaterials of variety of shapes can be prepared.

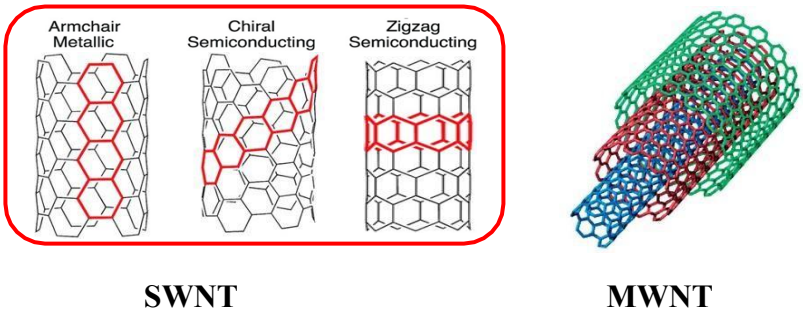
4.5 Carbon Nanomaterials:

Carbon nanotubes are a new form of a hexagonal network of carbon atoms rolled up in the form of cylindrical shape.

Types of CNTs:

1. **Single-Wall Nanotube (SWNT) (Arm Chair, Zig-Zag and Chiral)**
2. **Multi-Walled Nanotubes (MWNT)**

- Multiple rolled layers of graphene sheets



SWNT **MWNT**

Fig. 4.10: Types of carbon nanotubes

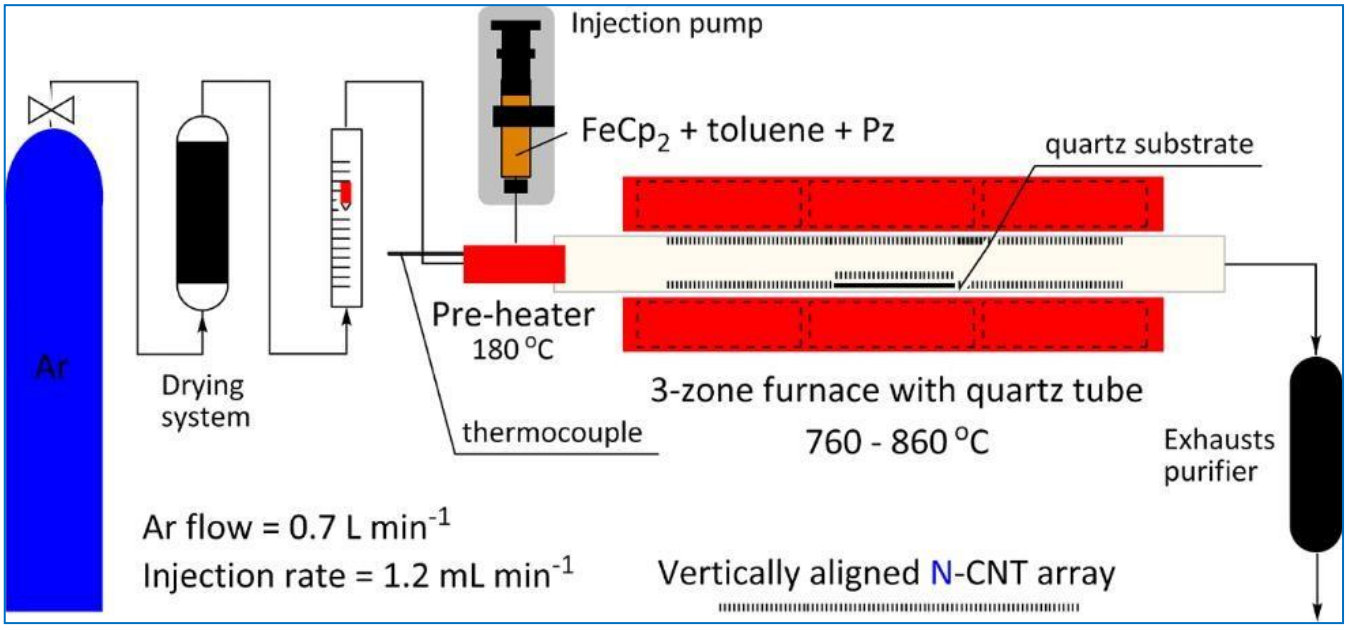
Synthesis of CNTs by modified CVD method:

Chemical Vapor Deposition: Requirements**:**

- High Temperature Tubular furnace (500 to 1500 ^{0}C)
- Source of Carbon: Benzene, xylene, natural gas, acetylene
- Substrate: carbon, quartz, silicon
- Inert gas: Argon, Hydrogen, Nitrogen
- Catalyst: Ferrocene, Nickellocene, Cobaltocene

In this technique, carbon nano tubes grow from the decomposition of hydrocarbons at temperature range of 500 to 1200 ^{0}C in the inert atmosphere on the catalyst activated substrate.

Fig. 4.10: CVD experimental set up (below)



Working: The above figure shows the schematic diagram of a typical catalytic chemical vapor deposition system. It is equipped with a horizontal tubular furnace as the reactor. The tube is made of quartz, 30 mm in diameter and 1000 mm in length. Ferrocene and Benzene/toluene vapors acts as the catalyst (Fe) and carbon atom precursors respectively were transported either by argon, hydrogen or mixture of both into the reaction chamber, and decomposed into respective ions of Fe and carbon atoms, resulting into carbon nanostructures. The growth of the nanostructures occurred in either the heating zone, before or after the heating zone, which is normally operated between 500ºC and 1150ºC for about 30 min.

Mechanism of CNTs growth:

1. **Diffusion** of Hydrocarbon gas
2. **Adsorption** of Hydrocarbon gas on the surface active site of catalyst particles.
3. **Dissociation** (Homolytic fission) of Hydrocarbon in to highly reactive free radicals of ‘C’ and ‘H’.
4. Repetition of steps 1, 2 and 3 leading to increase in concentration (saturation) of free radicals.

5. **Precipitation** (aggregation) of carbon free radicals results into crystal growth in the form tube on the surface. 'H' free radicals escape as H_2 gas.

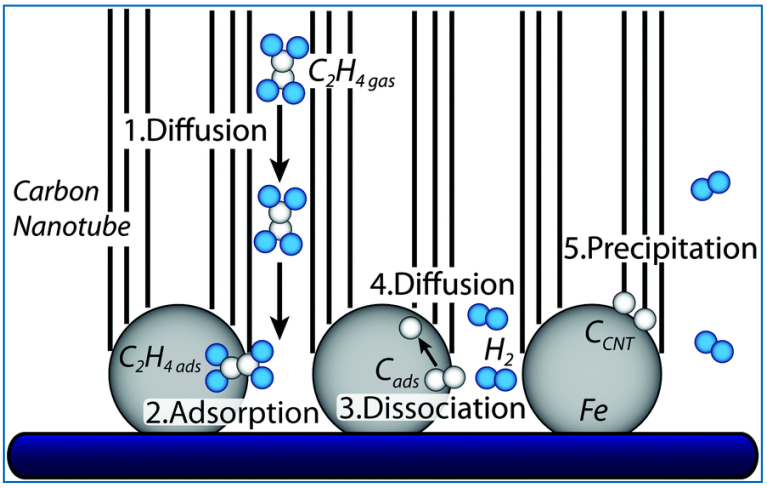
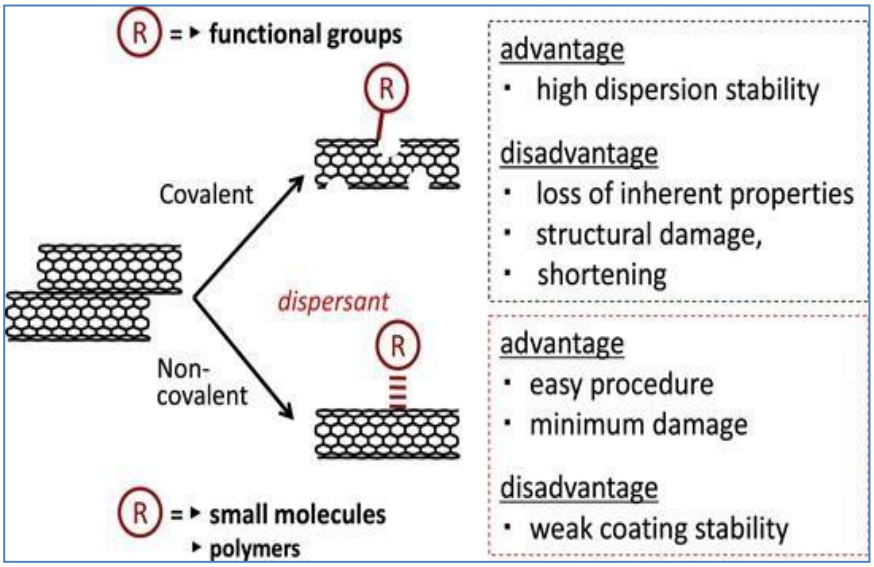


Fig. 4.11: Steps involved in CNTs growth

Functionalization of CNT:

CNTs may not have chemical affinity to the organic matrices and thus, the dispersion of CNTs into matrices is the biggest obstacle in practice. The modification/ functionalization of CNTs with other materials, makes it the most attractive and ultimate candidate for many applications which includes nanodevices, to organic electronics. The modification/ functionalization of CNTs can be simply divided into

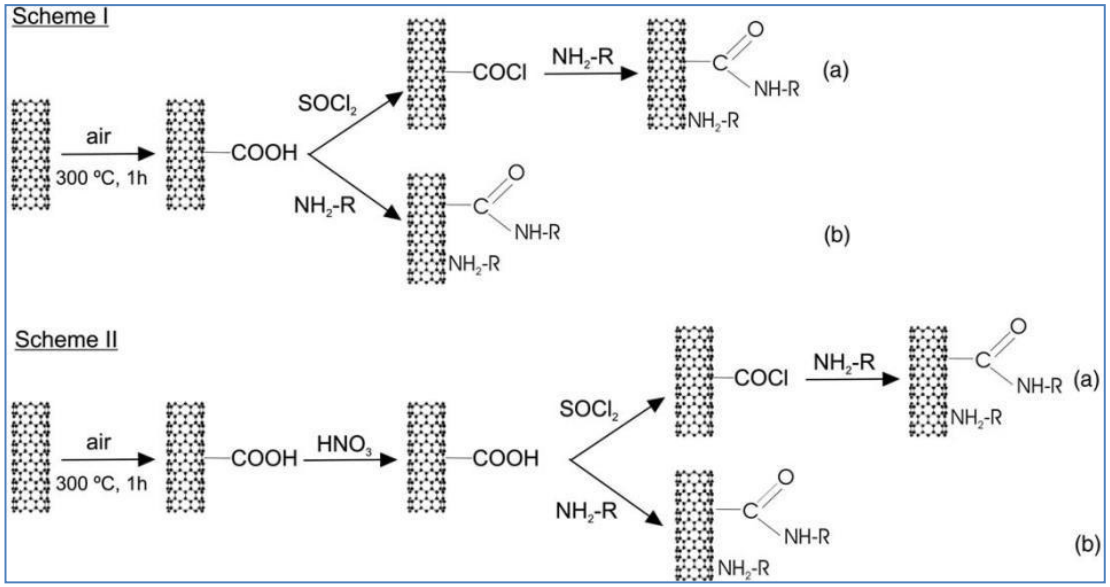
a) **Chemical (covalent) functionalization:** is based on the covalent bond of functional groups onto carbon form of CNTs. It can be performed at the end caps of nanotubes or at their sidewalls which have many defects. Direct covalent sidewall functionalization is associated with a change of hybridization from *sp2 to sp3 and a simultaneous loss of p-conjugation system.*



b) **Physical (Non-covalent) functionalization: It does not destroy the conjugated system** of the CNTs sidewalls, and therefore it does not affect the final structural properties of the material. The CNTs are functionalized non-covalently by **aromatic compounds, surfactants, and polymers,** employing ***π-π stacking or hydrophobic interactions*** for the most part. In this approaches, the non-covalent modifications of CNTs

can do much to preserve their desired properties, while improving their solubilities quite remarkably. (Ex: aromatic small molecule absorption, polymer wrapping, surfactants, biopolymers). Fig 12 : Types of CNT functionalization

Chemical (covalent) functionalization:



Physical (Non-covalent) functionalization:

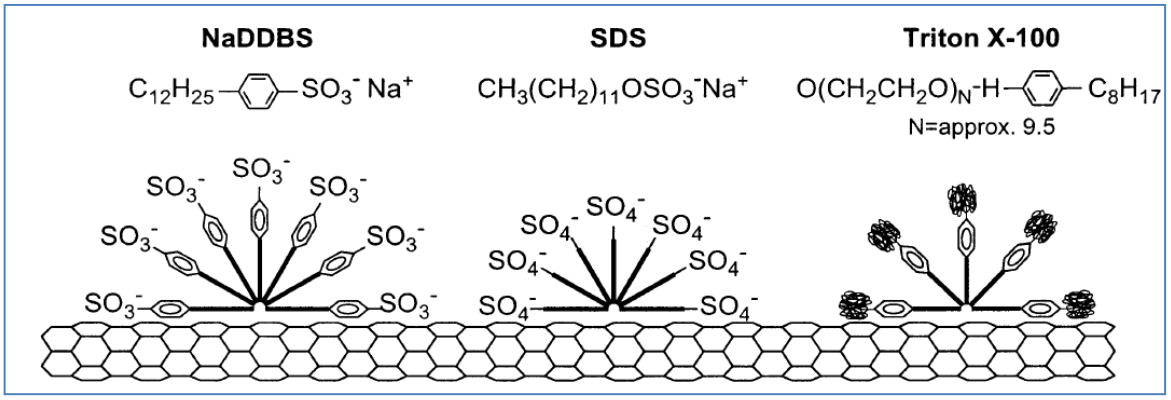
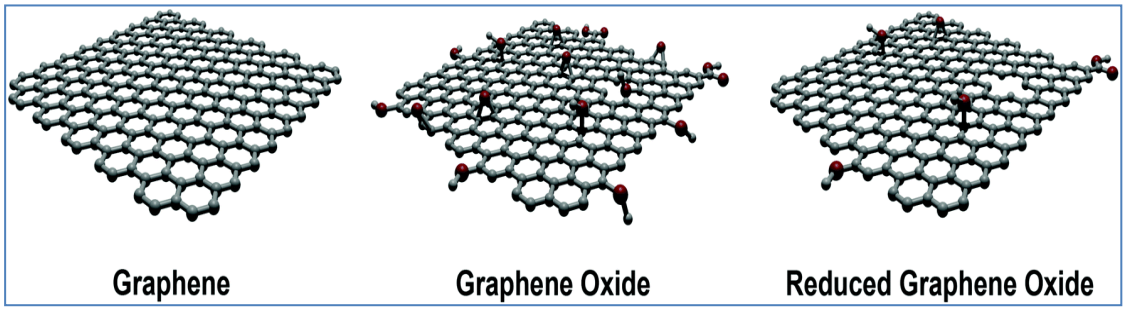


Fig. 4.12. Functionalization of Carbon nanotube

Applications of CNTs:

- Optoelectronic devices (Field Emission Display (FED))
- As *electrodes in* batteries, *capacitors and super capacitor electrodes*.
- As *electrode catalyst supports in Polymer Electrolyte Membrane (PEM) fuel cells*
- *Hydrogen* Storage material in hydrogen fuel car.
- As electrically *powered artificial* muscles.
- Electro catalyst for water splitting, H_2 production.

- Adsorbent and photocatalyst for organic pollutants degradation.
- Sensor for heavy metal ion, gases, volatile organic compounds and biomolecules.
- Drug carriers



4.6 Graphene: Graphene is an allotrope of carbon made of a single layer of carbon atoms that are bonded together in a repeating pattern of hexagons. It is a purely carbon-based, honeycomb-structured, one-atom thick layer of carbon atoms (two dimensional sheet), bonded to one another by sp2 hybridization.

Synthesis of Graphene Oxide and Reduced Graphene Oxide-Modified Hummer's Method:

In a typical procedure, graphite (2 g) and $NaNO_3$ (2 g) were combined with H_2SO_4 (90 ml) and stirred for 30 min in an ice bath. $KMnO_4$ (10 g) was added to the resulting solution, and the solution was then ultra sonicated at 50°C for 2 h. Deionized (DI) water (200 ml) and H_2O_2 (12 ml, 35%) were then slowly added to the above solution, and the resulting solution was washed with HCl (300 ml, 10%). Additional washing with concentrated HCl (200 ml, 37%) followed by drying gives GO powder. He formed GO is further subjected to reduction by reacting with sodium borohydride.

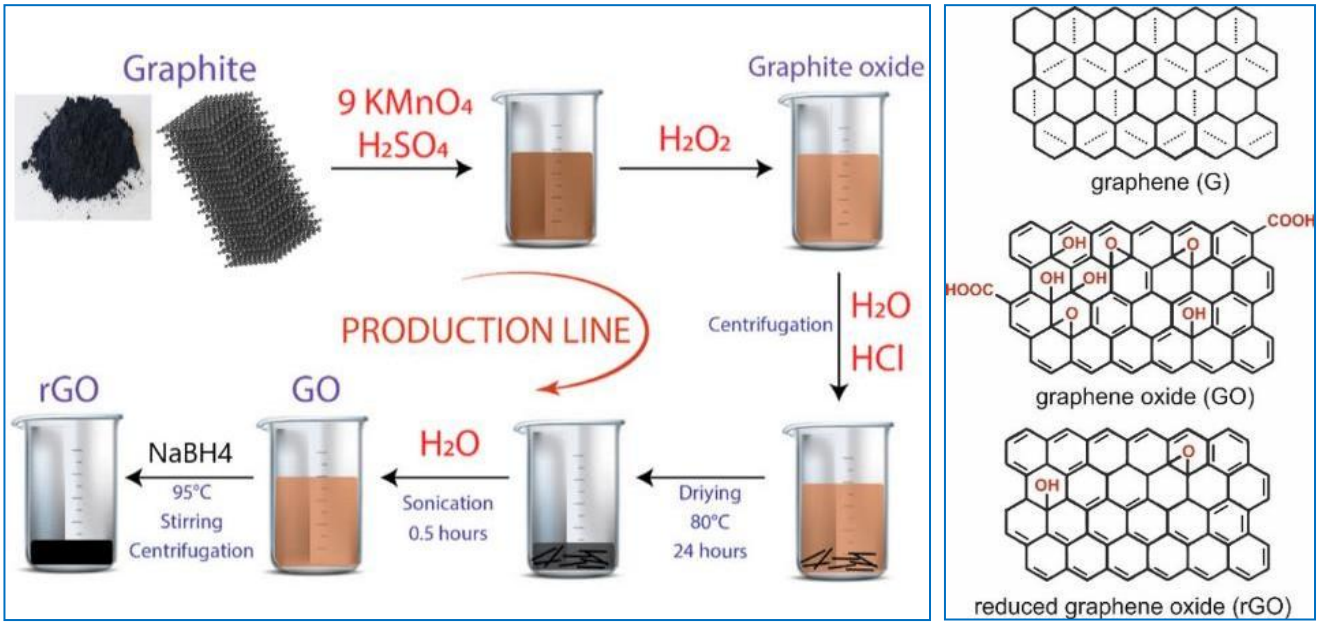
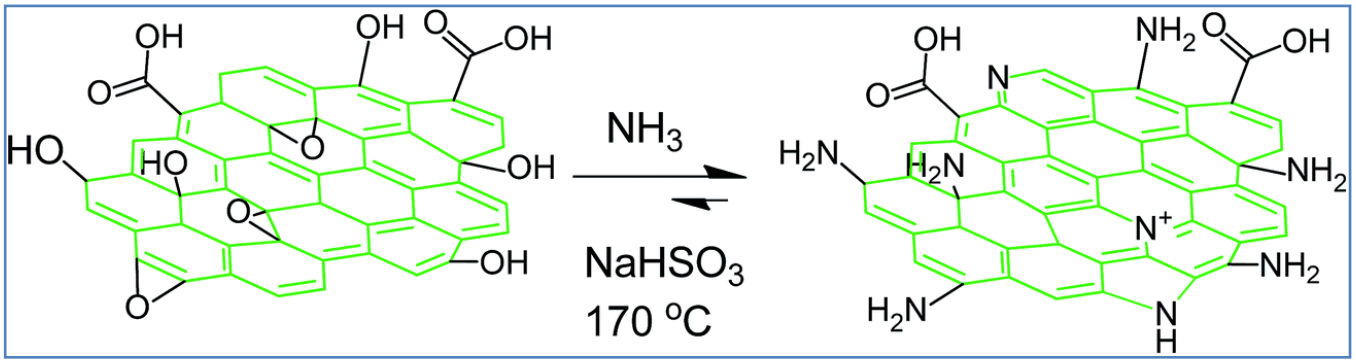


Fig 4.13: Modified hummers method steps

Functionalization of Graphene

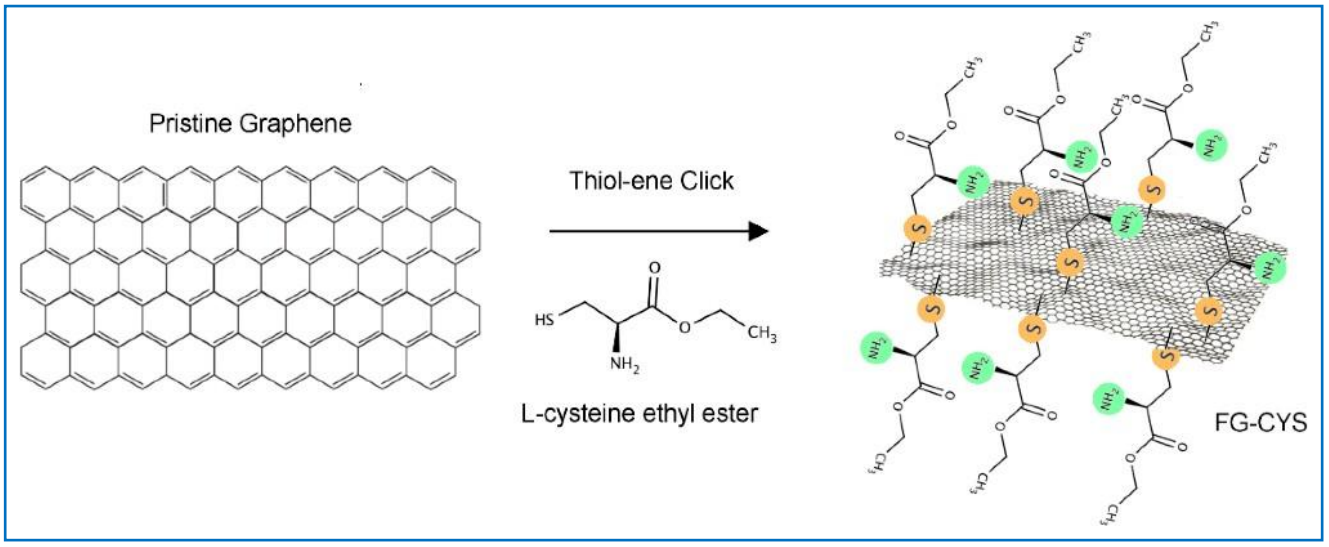
The functionalization is similar to CNT. It has two types (1) Covalent (chemical) and (2) Non-covalent (Physical) functionalization. The functionalization reduces the cohesive force between the graphene sheets and also to manipulate the physical and chemical properties.

The graphene oxide can be functionalized with amine group as follows.



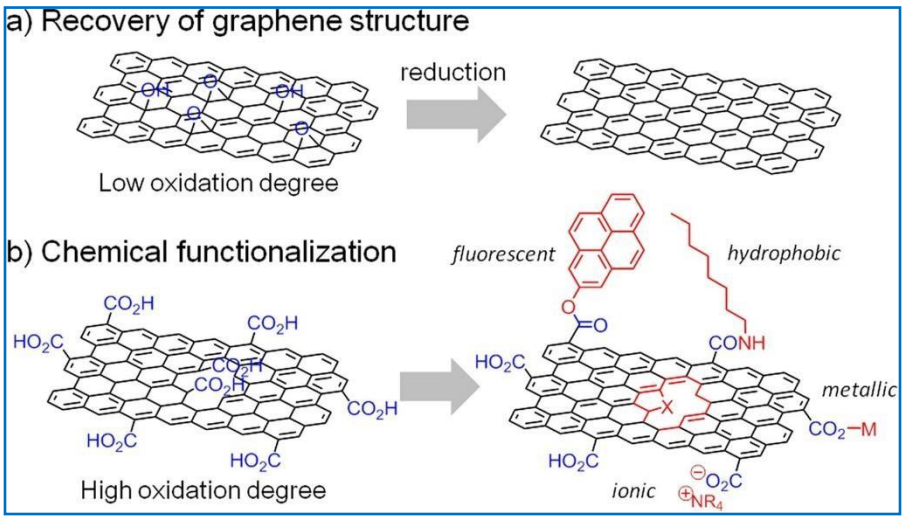
Ref: https://doi.org/10.1039/C5CC08252H

The graphene oxide can be functionalized with cysteine and thiolene group as follows.



Ref: https://doi.org/10.3390/ma14112830

Fig 4.14: Functionalization of Graphene



Ref: https://doi.org/10.1038/srep21715

Fig 4.15: Chemical functionalization of Graphene

Graphene Applications

Solar Cells, Batteries, Fuel cells, Medicine (Drug Delivery, Cancer Treatment, Gene Delivery, Photothermal Therapy, Diabetes Monitoring, Dialysis, Bone and Teeth Implantation, Tissue Engineering and Cell Therapy), UV Sensors, Biosensors, Transistors, Wearable Electronics, Flexible Screens, Optoelectronics, Optical Sensors, Water Purification, Desalination, Lubricants

Nano lubricants: *Types of nanoparticles as lubricant additives and their application in defence, automobile and spacecrafts.*

Nanolubricants have attracted great interest due to the promise of friction and wear reduction by introducing nanoparticles. It has great significance for energy conservation, emission reduction, and environmental protection. To date, the foremost challenge for developing a new nanolubricant is particle suspension. With the scientific and technological development, great advances have been made in nanolubricant additives in the scientific research and industrial applications.

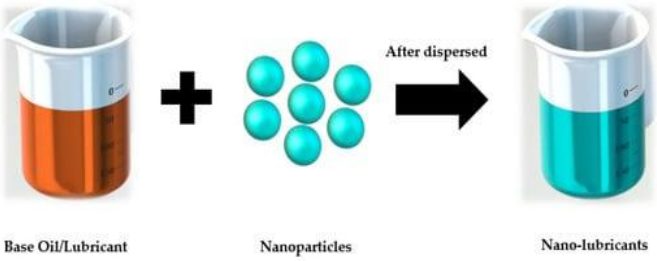


Fig 4.15: Nanolubricants composition

To reduce friction and wear, nanoparticles have been studied to be used as lubricant additives that have promising effects on friction and wear reduction in automotive, mining, and other industrial applications. Nanoparticles of various compositions and sizes have demonstrated certain degrees of friction modifying and anti-wear effects. Lubricants consisting of base oil and dispersed nanoparticles emerged as a new class of nanolubricants.

Nanomaterials as lubricant additives because of their small sizes are easy to enter friction contact areas and can form a protection tribofilm, thereby preventing the surface of friction pairs from being worn. In addition, nanomaterials have high surface activity, which can improve the film-forming stability of protection tribofilm by physical and/or chemical adsorption effects. There are three basic types of nanolubricant additives:

1. Nano metal-based additives,
2. Nano carbon-based additives,
3. Nano composite-based additives.

Table: 4.1. Types of nano-lubricant additives with examples.

Types of nano-lubricant additives		Examples
Nanometal-based lubricant additives	Pure metal	Cu, Ag, Fe, Pd, Ni
	Metal oxide	CuO, ZnO, Al_2O_3, TiO_2, ZrO_2
	Metal sulfide	WS_2, MoS_2, CuS, ZnS
	Metal hydroxide	$La(OH)_3$, LDHs
	Metal salt	$CaCO_3$, LaF_3, ZrP, Calcium Borate, Zinc Phosphate
Nanocarbon-based lubricant additives	Pure carbon	Nano Diamond, Fullerenes, Graphene, Carbon Nanotubes
	Polymer	PTFE, PSS, PVP
Nanocomposite-based lubricant additives		$Cu@SiO_2$, $Al_2O_3@TiO_2$, $Cu@MoS_2$, $G@MoS_2$, α-$Fe_2O_3@GO$, $FeS_2@G$, Ag@G, Cu@GO, $Mn_3O_4@G$, $La_2O_3@PI$, Alumina@MWCNT

Ref: *Jun ZHAO, Yiyao HUANG, Yongyong HE, Yijun SHI Friction* **9**(5): 891–917 (2021)

Lubrication mechanisms of nanolubricant additives

The lubrication mechanisms of nanoparticles can be briefly concluded into five aspects: formation of tribofilm, rolling bearing effect, the transformation of microstructure, synergistic effect, and surface repairing effect.

1. *Formation of tribofilm:* Because of the high values of specific surface area and surface energy, nanoparticles are easy to form a protective film on the contact surfaces by adsorption or chemical reactions, as shown in Fig. 10. The formation of tribofilm can avoid direct metal-to-metal contacts and play an important role in lubrication. The polarity of molecules significantly influences the film formation. Nanolubricant additives with low chemical activity such as copper, silver, and graphene always form a physical adsorption tribofilm.

2. *Rolling bearing effect:* Nanoparticles with spherical shapes can roll between the asperities of friction surfaces, which transforms sliding friction to rolling friction at a micro-area interface. Ex: SiO_2, WS_2 nanoparticles.

3. *Transformation of microstructure:* The microstructure is an important nature characteristic of nanoparticles such as carbon nanotubes, graphene, and nanodiamond. High pressure and heat during the friction process may destroy the original microstructure and produce a new microstructure. And the evolution of microstructure results in the variation of tribological performance. The microstructure of nanocarbon materials tends to be ordered during the friction process, thereby enhancing the antifriction and antiwear properties.

4. *Synergistic effect:* Nanolubricant additives are not isolated in lubricating oil or grease, and they often cooperate with other additives, lubricating oils, and modified friction surfaces to achieve a synergistic effect. Examples: Excellent tribological properties of silver nanoparticles were attributed to the synergistic effect between silver nanoparticles and base lubricants. Synergistic effect of Cu@rGO nanoparticles.

5. *Surface repairing effect*: The rubbing surfaces are generally smooth, of which the asperities can contact with each other directly at boundary lubrication, thereby leading to high friction coefficient and wear rate. Nanoparticles can repair the friction-induced surface defects and improve the tribological properties.

Parameters of Nanoparticles affect The Tribological Properties are 1. Concentration of Nanoparticles 2. Size of Nanoparticles, 3. Morphology of Nanoparticles, 4. Dispersion Stability of Nanoparticles

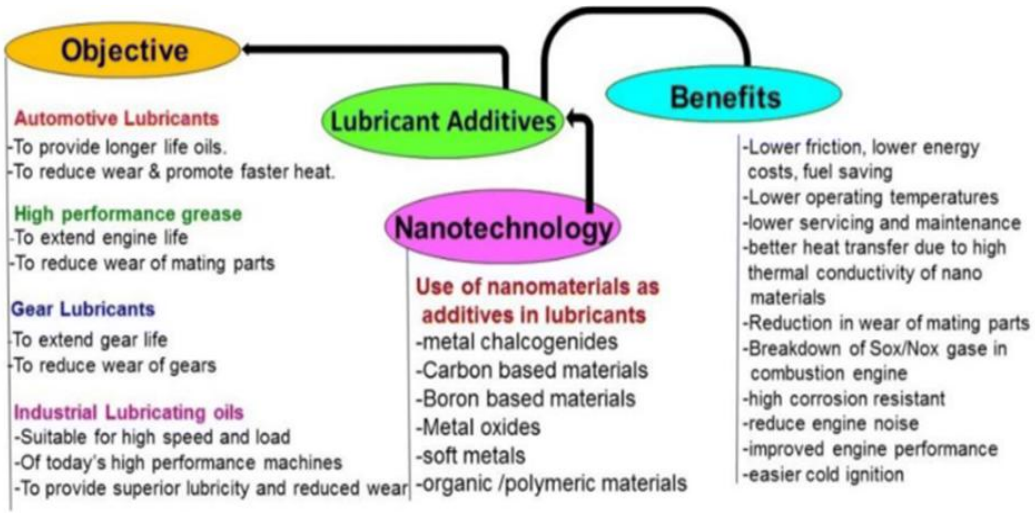


Fig 4.16: Nanolubricants advantages and Benefits

Application in automobile:

The wear and friction for the mechanical elements is a primary cause of energy dissipation in vehicle engines. The engine generated power is diminished in the range of 17-19% due to the frictional losses. Future automotive engines require more efficient engine oils, a state that appears a new challenge for designers and researchers worldwide regarding enhancing the vehicular engine tribology while achieving improving both the lube oil and fuel consumption.

The investigations on the fuel consumption reduction in automotive have become the essential research trend for different countries. Many explored the effect of Al2O3/TiO2 nano-lubricants on the behavior of the gasoline engine. The results explained that using nano-lubricants improves the engine behavior. Furthermore, the results revealed that the fuel consumption corresponding to Al2O3/TiO2 nano-lubricants declined by 16-20%.

Application in Spacecraft:

Lubricants are subject to potential degradation, depending on the amount of time they are used in various environments. Therefore, the number of days that a spacecraft will be in space, the number of duty cycles, the inorbit altitude, and the amount of time spent in ground storage are critical elements to consider in selecting a lubricant.

Lubricants are affected by the space environment. The ambient pressure can be as low as 10-12 torr, and the ambient temperature can be as low as -60°C, whereas the pressure and temperature inside the unit can range from 10-6 to 10-12 torr, and -50 to +20°C, respectively. This situation contributes to the loss of

lubricant by outgassing and a change in creep properties because of the change in viscosity, probably resulting in a reduced life time. The temperature of operation either increases or decreases the viscosity of a liquid lubricant, causing a change in the tendency of the lubricant to flow back into place between the contacting surfaces. The effect of a small temperature difference between two close surfaces (bearing inner and outer races) increases the tendency of an oil/grease lubricant to creep toward the cooler surface.

Application in Defense:

The use of high performance lubricating grease and transmission oil in the Armys ground and airborne vehicles can (i) enhance the performance and life of these systems, particularly in hot climates, (ii) reduce the heat signature, and (iii) help reduce the weight of thermal control systems, thereby enhancing the fuel efficiency of the vehicles.

END